

Task 2: Client Profitability and Spread Recommendation

The expected LP PnL per trade, assuming the position is closed entirely at horizon τ , is (Eq. 5):

$$E[\text{PnL}_\tau] = \text{side} \cdot V \cdot E[M_\tau - TP].$$

The aggregate expected PnL averages uniformly over all horizons (Eq. 6). A client is **profitable** if the aggregate PnL ≥ 0 at zero spread, and **costly** otherwise. The minimum half-spread is:

$$\delta_{\text{client}}^* = \max\left(0, \frac{-\overline{\text{PnL}}}{\bar{V}}\right).$$

Client	$\tau=5$	$\tau=10$	$\tau=15$	$\tau=20$	$\tau=25$	$\tau=30$	Agg. PnL	Class	δ^*
A	1.268	1.540	1.737	2.061	2.293	2.485	+1.898	Profitable	0.0000
B	1.211	1.343	1.542	1.726	1.909	2.059	+1.632	Profitable	0.0000
C	1.077	1.165	1.190	1.162	1.229	1.217	+1.173	Profitable	0.0000
D	0.868	0.682	0.493	0.177	-0.076	-0.347	+0.299	Profitable	0.0096
E	0.653	0.303	-0.108	-0.572	-1.069	-1.614	-0.401	Costly	0.0190
F	0.445	-0.176	-0.917	-1.751	-2.614	-3.629	-1.440	Costly	0.0329

Classification. Clients A, B, and C are **profitable**: their aggregate PnL is positive (+1.90, +1.63, +1.17 respectively) even at zero spread, and PnL *increases* with τ , indicating that mid-price movements over time favour the LP. Client D is marginally profitable overall (+0.30) but turns negative at $\tau \geq 25$ s, signalling a weak adverse-selection component at longer horizons. Clients E (-0.40) and F (-1.44) are **costly**: their aggregate PnL is negative and deteriorates sharply with τ , consistent with informed or directional flow that systematically moves the mid against the LP.

Relation of δ^* to the adversity profile. The adversity profile from Task 1 measures the frequency with which a client's trades turn adverse at each horizon. A steeply rising adversity curve implies that, as τ grows, an increasing fraction of trades produce a negative mark-to-market for the LP — which directly depresses the aggregate PnL. Because δ^* is the charge needed to recover that deficit, there is a direct monotone link: clients with higher and faster-rising adversity require a larger δ^* . Concretely, Clients A–C have flat adversity ($\leq 46\%$ at all τ) and $\delta^* = 0$; Client D's moderately rising profile ($43\% \rightarrow 52\%$) yields $\delta^* \approx 0.0096$; and Clients E and F, with the steepest adversity curves ($45\% \rightarrow 55\%$ and $47\% \rightarrow 62\%$), require $\delta^* \approx 0.019$ and 0.033 respectively. In short, the *slope* and *level* of the adversity curve are the primary drivers of the minimum viable spread.

